



CALCULO INTEGRAL BMA02

semana 3

Método de integración: De racionales

$$I = \int \frac{P(x)}{Q(x)} dx$$





Método de Integración de racionales

$$I = \int \frac{P(x)}{Q(x)} dx \quad \text{Fracciones propias}$$

caso I) factorizando el denominador, los factores primos son lineales

$$I = \int \frac{P(x)}{(x-a)(x-b)(x-c)} dx = \int \frac{A}{(x-a)} dx + \int \frac{B}{(x-b)} dx + \int \frac{C}{(x-c)} dx$$

$$\frac{P(x)}{(x-a)(x-b)(x-c)} = \frac{A}{(x-a)} + \frac{B}{(x-b)} + \frac{C}{(x-c)}$$

ejemplo: $I = \int \frac{2x-1}{x^3+x^2-4x-4} dx$

$$\frac{2x-1}{x^3+x^2-4x-4} = \frac{2x-1}{(x+1)(x+2)(x-2)}$$



$$I = \int \frac{dx}{x+1} - \frac{5}{4} \int \frac{dx}{x+2} + \frac{1}{4} \int \frac{dx}{x-2} = \ln|x+1| - \frac{5}{4} \ln|x+2| + \frac{1}{4} \ln|x-2| + C$$



Método de Integración de racionales

$$I = \int \frac{P(x)}{Q(x)} dx \quad \text{Fracciones propias}$$

caso II) factorizando el denominador, los factores primos son lineales y algunos se repiten.

$$I = \int \frac{P(x)}{(x-a)(x-b)^3} dx = \int \frac{A}{(x-a)} dx + \int \frac{B}{(x-b)} dx + \int \frac{C}{(x-b)^2} dx + \int \frac{D}{(x-b)^3} dx$$

$$\frac{P(x)}{(x-a)(x-b)^3} = \frac{A}{(x-a)} + \frac{B}{(x-b)} + \frac{C}{(x-b)^2} + \frac{D}{(x-b)^3}$$

ejemplo: $I = \int \frac{4x^2 - 10x + 12}{x^3 - 2x^2 - 4x + 8} dx$

$$\frac{4x^2 - 10x + 12}{x^3 - 2x^2 - 4x + 8} = \frac{4x^2 - 10x + 12}{(x-2)^2(x+2)}$$

$$I = \int \frac{dx}{x-2} + 2 \int \frac{dx}{(x-2)^2} + 3 \int \frac{dx}{x+2} = \ln|x-1| - \frac{2}{x-2} + 3\ln|x+2| + C$$



Método de Integración de racionales

$$I = \int \frac{P(x)}{Q(x)} dx \quad \text{Fracciones propias}$$

caso III) factorizando el denominador, existen factores primos cuadráticos y no se repite ninguno.

$$I = \int \frac{P(x)}{(ax^2+bx+c)(dx+f)} dx = \int \frac{Ax+B}{ax^2+bx+c} dx + \int \frac{C}{dx+f} dx \quad \frac{B}{(x-b)} +$$

$$\frac{P(x)}{(ax^2+bx+c)(dx+f)} = \frac{Ax+B}{(ax^2+bx+c)} + \frac{C}{dx+f}$$

$$\text{ejemplo: } I = \int \frac{1}{(x^2+3x+1)(x-1)} dx \quad \frac{1}{(x^2+3x+1)(x-1)} = \frac{Ax+B}{(x^2+3x+1)} + \frac{C}{x-1}$$

$$I = -\frac{1}{5} \int \frac{x+4}{(x^2+3x+1)} + \frac{1}{5} \int \frac{dx}{x-1} = -\frac{1}{10} \ln|x^2+3x+1| - \frac{1}{\sqrt{5}} \ln \left| \frac{2x+3+\sqrt{5}}{2\sqrt{x^2+3x+1}} \right| + C$$



Método de Integración de racionales

Calcule: $I = \int \frac{dx}{x[\operatorname{Ln} x - 1]^3[\operatorname{Ln} x + 2]^4}$

$$u = \operatorname{Ln} x \Rightarrow du = \frac{dx}{x} \Rightarrow I = \int \frac{du}{[u - 1]^3[u + 2]^4} \quad \text{sea } t = \frac{u + 2}{u - 1}$$

$$u = \frac{t + 2}{t - 1} \quad u - 1 = \frac{3}{t - 1} \quad u + 2 = \frac{3t}{t - 1} \quad du = \frac{-3dt}{(t - 1)^2}$$

$$I = -3 \int \frac{dt}{[t - 1]^2 \left[\frac{3}{t - 1} \right]^3 \left[\frac{3t}{t - 1} \right]^4} = -\frac{1}{729} \int \frac{[t - 1]^5 dt}{t^4} = -\frac{1}{729} \int \frac{(t^5 - 5t^4 + 10t^3 - 10t^2 + 5t - 1) dt}{t^4}$$

$$I = -\frac{1}{729} \int (t^4 - 5 + 10t^{-1} - 10t^{-2} + 5t^{-3} - t^{-4}) dt = -\frac{1}{729} \left[\frac{t^5}{5} - 5t + 10\operatorname{Ln}|t| + 10t^{-1} - \frac{5}{2}t^{-2} + \frac{1}{3}t^{-3} \right] + C$$

$$I = -\frac{1}{729} \left[\frac{\left(\frac{\operatorname{Ln} x + 2}{\operatorname{Ln} x - 1} \right)^5}{5} - 5 \left(\frac{\operatorname{Ln} x + 2}{\operatorname{Ln} x - 1} \right) + 10\operatorname{Ln} \left| \frac{\operatorname{Ln} x + 2}{\operatorname{Ln} x - 1} \right| + 10 \left(\frac{\operatorname{Ln} x - 1}{\operatorname{Ln} x + 2} \right) - \frac{5}{2} \left(\frac{\operatorname{Ln} x - 1}{\operatorname{Ln} x + 2} \right)^2 + \frac{1}{3} \left(\frac{\operatorname{Ln} x - 1}{\operatorname{Ln} x + 2} \right)^3 \right] + C$$



PROBLEMAS PARA LA PC1

1. - determine : $I = \int (1 + \ln^2 x)^2 (\arctan(1 + \ln^2 x)) \left(\frac{\ln x}{x} \right)$

$$1 + \ln^2 x = t \Rightarrow \frac{2 \ln x}{x} \cdot dx = dt \quad \rightarrow \quad I = \frac{1}{2} \int t^2 \arctan(t) \cdot dt = \frac{1}{2} \left(\frac{1}{3} \right) \int \boxed{\arctan(t)} \cdot d(\boxed{t^3})$$

$\uparrow u$ $\uparrow v$

$$I = \frac{1}{6} t^3 \arctan t - \frac{1}{6} \int \boxed{\frac{t^3}{t^2 + 1}} dt \quad \xrightarrow{I_1} \quad I_1 = \int \frac{t^3}{t^2 + 1} dt = \int \left(t - \frac{t}{t^2 + 1} \right) dt$$

$$I_1 = \int t dt - \frac{1}{2} \int \frac{2t}{t^2 + 1} dt = \frac{t^2}{2} - \frac{1}{2} \ln|t^2 + 1| \quad \rightarrow \quad I = \frac{1}{6} t^3 \arctan t - \frac{t^2 - \ln(t^2 + 1)}{12} + C$$

$$I = \frac{2(1 + \ln^2 x)^3 \arctan(1 + \ln^2 x) - (1 + \ln^2 x)^2 + \ln((1 + \ln^2 x)^2 + 1)}{12} + C$$



PROBLEMAS PARA LA PC1

2. – Calcule: $I = \int \frac{e^{e^{\ln(\cos 2x) - \ln(\sec 2x)}} + e^{\cos(2x) + 4\ln(x^{\frac{1}{4}} \sin(2x))}}{e^{(\cos(2x) - 3 \ln(\csc(2x)))}} dx$



PROBLEMAS PARA LA PC1

$$3.- I = \int \frac{(x^4 + 4x^3 + 11x^2 + 12x + 8)}{(x^2 + 2x + 3)^2(x + 1)} dx \quad x + 1 = u$$

$$\frac{(x^4 + 4x^3 + 11x^2 + 12x + 8)}{(x^2 + 2x + 3)^2(x + 1)} = \frac{(x + 1)^4 + 5(x^2 + 2x + 1) - 2(x + 1) + 4}{(x + 1)((x + 1)^2 + 2)^2} +$$

$$I = \int \frac{u^4 + 5u^2 - 2u + 4}{(u^2 + 2)^2 u} du \quad \frac{u^4 + 5u^2 - 2u + 4}{(u^2 + 2)^2 u} = \frac{A}{u} + \frac{Bu + C}{(u^2 + 2)} + \frac{Du + E}{(u^2 + 2)^2}$$

$$u^4 + 5u^2 - 2u + 4 = A((u^2 + 2)^2) + (Bu + C)u(u^2 + 2) + (Du + E)u \quad A = 1$$

$$4 + 5(-2) - 2u + 4 = (D(-2) + Eu) \quad B = C = 0 \quad E = -2 \quad D = 1$$

$$I = \int \frac{du}{u} + \frac{1}{2} \int \frac{2u - 4}{(u^2 + 2)^2} du = \ln|u| + \frac{1}{2} \int \frac{2u}{(u^2 + 2)^2} du - 2 \int \frac{1}{(u^2 + 2)^2} du$$



PROBLEMAS PARA LA PC1

$$3.- I = \int \frac{(x^4 + 4x^3 + 11x^2 + 12x + 8)}{(x^2 + 2x + 3)^2(x + 1)} dx \quad x + 1 = u$$

$$I = \ln|u| + \frac{1}{2} \int \frac{2u}{(u^2 + 2)^2} du - 2 \int \frac{1}{(u^2 + 2)^2} du \quad I_1 = \int \frac{1}{(u^2 + 2)^2} du$$

$$u = \sqrt{2} \tan \theta \quad du = \sqrt{2} \sec^2 \theta d\theta \quad I_1 = \int \frac{\sqrt{2} \sec^2 \theta}{4 \sec^4 \theta} d\theta = \frac{\sqrt{2}}{4} \int \cos^2 \theta d\theta$$

$$I_1 = \frac{\sqrt{2}}{8} \int (1 + \cos 2\theta) d\theta = \frac{\sqrt{2}}{8} \left(\theta + \frac{\sin 2\theta}{2} \right)$$

$$I = \ln|u| - \frac{1}{2(u^2 + 2)} - \frac{\sqrt{2}}{4} \arctan \frac{u}{\sqrt{2}} + \frac{u}{2(u^2 + 2)} + C$$

$$I = \ln|x + 1| - \frac{1}{2(x^2 + 2x + 3)} - \frac{\sqrt{2}}{4} \arctan \frac{x + 1}{\sqrt{2}} + \frac{x + 1}{2(x^2 + 2x + 3)} + C$$



PROBLEMAS PARA LA PC1

4. – Calcule: $I = \int \frac{e^{\ln(\ln x)^2} + \ln x + 2}{x(\ln x)^3 + 8x} dx$ $\ln x = u$ $\frac{dx}{x} = du$

$$I = \int \frac{u^2 + u + 2}{u^3 + 8} du = \int \frac{u^2 + u + 2}{(u + 2)(u^2 + 2u + 4)} du = \int \frac{A}{(u + 2)} du + \int \frac{Bu + C}{(u^2 + 2u + 4)} du$$

$$u^2 + u + 2 = A(u^2 + 2u + 4) + (Bu + C)(u + 2) \quad A = 1 \quad B = 0 \quad C = -1$$

$$I = \int \frac{1}{(u + 2)} du - \int \frac{1}{(u^2 + 2u + 4)} du = \ln|u + 2| - \frac{1}{\sqrt{3}} \arctan\left(\frac{u + 1}{\sqrt{3}}\right) + C$$

$$I = \ln|\ln x + 2| - \frac{1}{\sqrt{3}} \arctan\left(\frac{\ln x + 1}{\sqrt{3}}\right) + C$$



PROBLEMAS PARA LA PC1

5. -Calcule: $I = \int \frac{e^x(1 - 5x + x^2 + \sin x - 5x \cos x + x^2 \cos x) dx}{1 + \cos x}$

$$I = \int \frac{e^x(1 - 5x(1 + \cos x) + x^2(1 + \cos x) + \sin x) dx}{1 + \cos x}$$

$$I = \int e^x x^2 dx - 5 \int x e^x dx + \int \frac{e^x(1 + \sin x) dx}{1 + \cos x} \quad I1 = \int e^x x^2 dx = e^x x^2 - 2 \int x e^x dx$$

$$I = e^x x^2 - 7x e^x + 7 + \int \frac{e^x(1 + \sin x) dx}{1 + \cos x}$$


$$\int \left(\frac{1 + \sin x}{1 + \cos x} \right) e^x dx \Rightarrow \int \frac{e^x}{1 + \cos x} dx + \int \frac{e^x \sin x}{1 + \cos x} dx$$



PROBLEMAS PARA LA PC1

$$I = e^x x^2 - 7xe^x + 7 + \int \frac{e^x(1 + \sin x) dx}{1 + \cos x}$$

$$\int \left(\frac{1 + \sin x}{1 + \cos x} \right) e^x dx \Rightarrow \int \frac{e^x}{1 + \cos x} dx + \int \frac{e^x \sin x}{1 + \cos x} dx$$

$$I_2 = \int \frac{e^x}{2\cos^2\left(\frac{x}{2}\right)} dx + \int \frac{2e^x \sin\left(\frac{x}{2}\right) \cos\left(\frac{x}{2}\right)}{2\cos^2\left(\frac{x}{2}\right)} dx$$

$$I_2 = e^x \tan\left(\frac{x}{2}\right) - \int e^x \tan\left(\frac{x}{2}\right) dx + \int e^x \tan\left(\frac{x}{2}\right) dx dx$$

$$I = e^x x^2 - 7xe^x + 7 + e^x \tan\left(\frac{x}{2}\right)$$



PROBLEMAS PARA LA PC1

$$6. - I = \int \frac{[(\arcsen(x)) \ln(\arcsen(x))]^2}{\sqrt{1-x^2}} dx$$





Método de Integración de potencias seno y coseno

Integrales de la forma: $\int \text{sen}^n x dx$, $\int \text{cos}^m x dx$

i) Para n , par Usar identidades trigonométricas

$$\text{sen}^2 x = \frac{1 - \cos 2x}{2}$$

$$\text{cos}^2 x = \frac{1 + \cos 2x}{2}$$

ii) Para n , impar Usar las identidades pitagóricas

$$\text{sen}^2 x + \text{cos}^2 x = 1$$

Ejemplo 1 Obtenga: $I = \int \text{cos}^7 x dx$ $I = \int [\text{cos}^2 x]^3 \text{cos} x dx$

$$I = \int [1 - \text{sen}^2 x]^3 d\text{sen} x = \int [1 - 3\text{sen}^2 x + 3\text{sen}^4 x - \text{sen}^6 x] d\text{sen} x$$

$$I = \text{sen} x - \text{sen}^3 x + \frac{3}{5} \text{sen}^5 x - \frac{1}{7} \text{sen}^7 x + C$$



Método de Integración de potencias seno y coseno

Ejemplo 2 Obtenga: $I = \int \text{sen}^6 x dx$ $I = \int \left[\frac{1 - \cos 2x}{2} \right]^3 dx$

$$I = \frac{1}{8} \int [1 - 3\cos 2x + 3\cos^2 2x - \cos^3 2x] dx$$

$$I = \frac{1}{8} \int \left[1 - 3\cos 2x + 3 \left[\frac{1 + \cos 4x}{2} \right] - \cos^2 2x \cos 2x \right] dx$$

$$I = \frac{1}{8} \int \left[1 - 3\cos 2x + \frac{3}{2} + \frac{3}{2} \cos 4x \right] dx - \frac{1}{16} \int [1 - \text{sen}^2 2x] d\text{sen} 2x$$

$$I = \frac{5}{16} x - \frac{1}{4} \text{sen} 2x + \frac{3}{64} \text{sen} 4x - \frac{1}{48} \text{sen}^3 2x + C$$



Método de Integración de potencias seno y coseno

Integrales de la forma: $\int \text{sen}^n x \cos^m x dx$

i) Uno de los exponentes es impar, cambio $u = \cos x$, cuando m es par

Ejemplo 3 Obtenga: $I = \int \text{sen}^3 x \cos^{-4} x dx$ $u = \cos x \rightarrow du = -\text{sen} x dx$

$$I = \int \text{sen}^2 x \text{sen} x \cos^{-4} x dx = - \int (1 - \cos^2 x) \cos^{-4} x du = - \int (1 - u^2) u^{-4} du$$

$$I = \int (u^{-2} - u^{-4}) du = \frac{1}{3} \cos^{-3} x - \cos^{-1} x + C = \frac{1}{3} \sec^3 x - \sec x + C$$

Ejemplo 4 Obtenga: $I = \int \text{sen}^3 x \cos^5 x dx = \int \text{sen}^3 x \cos^4 x \cos x dx$

$$I = \int \text{sen}^3 x [1 - \text{sen}^2 x]^2 x d\text{sen} x = \int \text{sen}^3 x [1 - 2\text{sen}^2 x + \text{sen}^4 x] d\text{sen} x$$

$$I = \frac{1}{4} \text{sen}^4 x - \frac{1}{3} \text{sen}^6 x + \frac{1}{8} \text{sen}^8 x + C$$



Método de Integración de potencias seno y coseno

Integrales de la forma: $\int \text{sen}^n x \cos^m x dx$

ii) Los dos exponentes son pares, use angulo doble

Ejemplo 5 Obtenga: $I = \int \text{sen}^2 x \cos^4 x dx = \int \text{sen}^2 x (\cos^2 x)^2 dx$

$$I = \int \frac{1 - \cos 2x}{2} \left[\frac{1 + \cos 2x}{2} \right]^2 dx = \frac{1}{8} \int [1 - \cos^2 2x][1 + \cos 2x] dx$$

$$I = \frac{1}{8} \int [1 + \cos 2x - \cos^3 2x - \cos^2 2x] dx$$

$$I = \frac{1}{8} \int \left[1 + \cos 2x - \frac{1}{2} (1 - \text{sen}^2 2x) 2 \cos 2x - \frac{1 + \cos 4x}{2} \right] dx$$

$$I = \frac{1}{8} \left[\frac{x}{2} + \frac{1}{6} \text{sen}^3 2x - \frac{1}{8} \text{sen} 4x \right] + C$$